ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 2 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (practical, 100 marks): a Scratch recycling-points app with an IPO chart, a Python turtle drawing of coloured squares, and a Python school-transport fee program with a function, selection, a discount, a loop and a global total.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Scratch for QUESTION 1 and a Python IDE (Python 3) for QUESTIONS 2 and 3. If you do not have a computer, write the programs neatly in your answer book.
3. Save each program with the question number and your name or examination number. Add your name or examination number as a comment on the first line of each Python program.
4. Save your work regularly. Programs are not marked strictly on syntax.

QUESTION 1

A school runs a recycling competition. Learners bring plastic, glass and paper, and earn points: 3 points per kg of plastic, 2 points per kg of glass and 1 point per kg of paper. A learner with 50 points or more earns a GOLD badge, 25 to 49 points a SILVER badge, and fewer than 25 points a BRONZE badge. Create a Scratch app in which the sprite asks for the mass of each material, works out the points and says the badge.

INPUT	PROCESS	OUTPUT
kg of plastic kg of glass kg of paper	Declare all variables and set them to 0 Ask for the three masses points = plastic × 3 + glass × 2 + paper × 1 If points ≥ 50: badge = "GOLD" Else if points ≥ 25: badge = "SILVER" Else: badge = "BRONZE"	points badge

1.1 Open Scratch, create a new project and name it Question1_YourName. (1)

1.2 Choose a sprite to act as the competition judge. (1)

1.3 Choose a suitable backdrop (for example a school or outdoors). (1)

1.4 Declare all the variables and give them their starting values. (8)

1.5 Ask for the mass of plastic, glass and paper, and store each answer. (6)

1.6 Calculate the points. (4)

1.7 Use if … then … else blocks to choose the badge. (8)

1.8 Display the points and the badge. (3)

1.9 Test the program with these values and write down what the sprite says: 10 kg plastic, 5 kg glass, 8 kg paper; 20 kg plastic, 0 kg glass, 0 kg paper; 2 kg plastic, 3 kg glass, 4 kg paper. (3)

[35]

QUESTION 2

Write a Python program that uses the turtle graphics library to draw a row of EIGHT squares, side by side, each with sides of 40 pixels. The squares must be filled in alternating colours: red, blue, red, blue …



FIGURE 2.1: The row of squares

2.1 Create a new file called Question2_YourName.py and add your name or examination number as a comment on the first line. (1)

2.2 Import the turtle module and create a turtle object named builder. (2)

2.3 Set the drawing speed to 0. (2)

2.4 Set the thickness of the lines to 2 pixels. (2)

2.5 Use a loop that draws the square 8 times, moving the turtle 40 pixels to the right before each new square. (5)

2.6 Use a second (inner) loop to draw each square: 4 sides of 40 pixels with a 90° turn after each side. (5)

2.7 Fill the squares in alternating colours (red for the 1st, 3rd, 5th … square and blue for the others), using an if statement. (5)

[22]

QUESTION 3

A school organises bus transport for learners. Learners who live less than 5 km from the school walk and pay nothing. Learners who live from 5 km up to 15 km pay R150 per month, and learners who live more than 15 km away pay R250 per month. A learner who has a brother or sister on the bus gets a 20% discount. The school wants a Python program that registers learners one by one and shows the total monthly fees at the end.

The following steps show how the program should be used:

```
Learner name: Ayanda
Distance from school (km): 8
Sibling on the bus? (yes/no): no
Ayanda pays R150.00 per month
Register another learner? (yes/no): yes
Learner name: Kabelo
Distance from school (km): 20
Sibling on the bus? (yes/no): yes
Kabelo pays R200.00 per month (20% sibling discount)
Register another learner? (yes/no): no
Total monthly fees: R350.00
```

Follow the instructions below to create the application:

3.1 Create a new Python file called Question3_YourName.py. (1)

3.2 Declare a global variable named total_fees and set it to 0. (2)

3.3	Create a function named register_learner() with no parameters. Inside it, ask for the learner's name, the distance from school and whether a sibling is on the bus.	(6)
3.4	Use if, elif and else to choose the monthly fee from the distance.	(8)
3.5	Give a 20% discount when the learner has a sibling on the bus.	(4)
3.6	Display the fee for the learner.	(4)
3.7	Use a loop that keeps registering learners until the answer is "no".	(7)
3.8	Add each learner's fee to total_fees.	(4)
3.9	Call the function inside the loop and display the total monthly fees at the end.	(4)
3.10	Test the program with the data shown in the example above.	(2)
3.11	Save the program and print (or neatly present) the code.	(1)
		[43]
		TOTAL: 100

END OF QUESTION PAPER